Stereochemistry

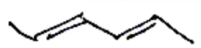
- Isomers
  - Structural
  - Stereoisomer
    - Configurational
      - Optical (enantiomer, diastereomer)
      - Geometrical
    - Conformational

Geometrical Isomerism :

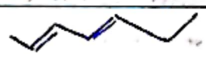
Occurs due to restricted rotation. High amount of energy req $> 200^\circ c$
The pi-bond gets disturbed due to this.
Eg: cyclic systems, $C=C$, $C=N$, $N=N$.

↳ CIP Rules :

(1) Atomic Number. (2) Atomic Mass.
(3) Increase depth search one-by-one.

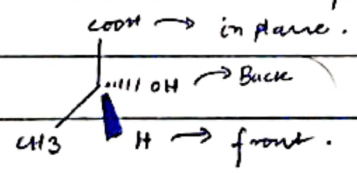
Symmetrical : 
$n \rightarrow$ no. of double bond

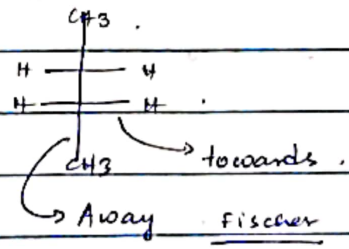
even : $GI = 2^{n-1} + 2^{n/2 - 1}$
odd : $GI = 2^{n-1} + 2^{n-1/2}$

Unsymmetrical  : $GI = 2^n$.

Optical Isomerism :

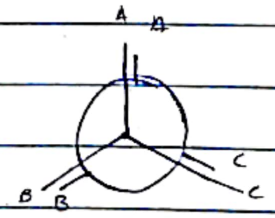
Enantiomer : Non super-imposable mirror image.





Flying Wedge. Fischer

Newmann Projection : see molecule front front view



↳ If multiple axis have same C_n, then choose max n as principal axis otherwise choose the one which passes through most atoms.

apsara

Date: __.__.__

↳ Plane symmetry (σ)

If passing mirror through an axis, if molecule is cut into two equal halves → POS exists.

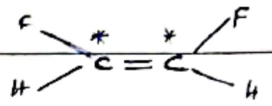
vertical POS (σ_v) } wrt principal axis → parallel

horizontal POS (σ_h) } wrt principal axis → perpendicular.

↳ Inversion Centre : equal distance - same atoms about COS (i)

↳ Chirality : centre with all different groups. (chiral centre)

↳ Stereocentre : changing position of atoms there changes the molecule.



laevorotatory (−) l. } observed rotation (θ)

dextrotatory (+) d. } observed rotation (θ)

↳ same molecule should give same amount of rotation. $[\alpha]$ → specific

⇒ $\boxed{\theta = [\alpha] \cdot l \cdot c}$ → concentration

↳ length of tube.

1) Molecule should contain chiral C. } optical activity check ?

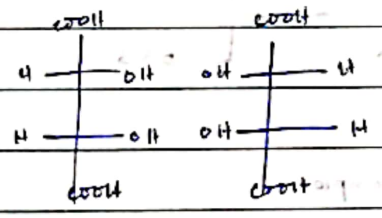
2) Molecule shouldn't have POS } optical activity check ?

■ Enantiomers :

Non-superimposable mirror images.

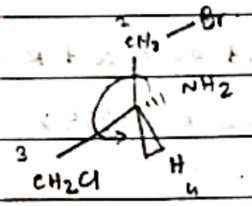
■ Diastereomers :

Non-mirror, non-superimposable.

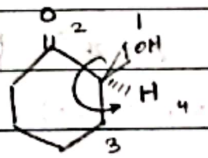


These two are bromomers.

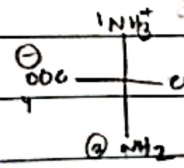
Check by rotating 180°. Since optically inactive.

Q. 

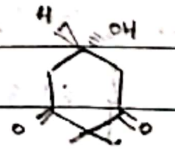
ACW (front) (R)

Q. 

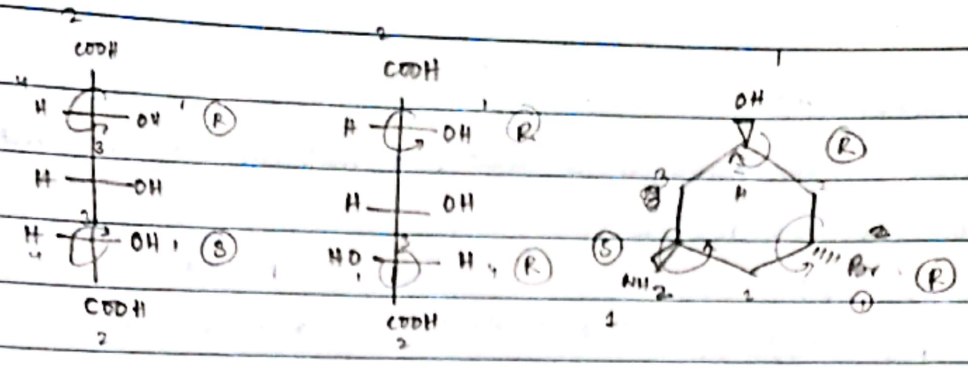
ACW (back) (S)

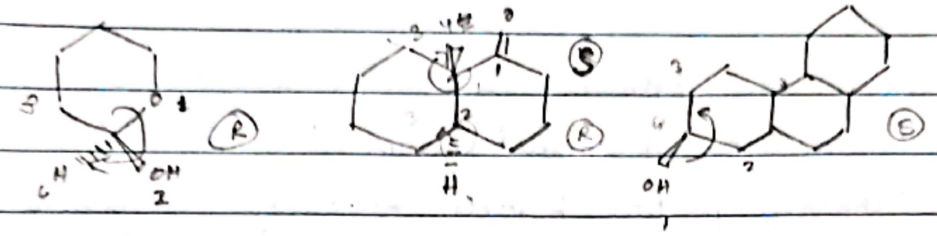
Q. 

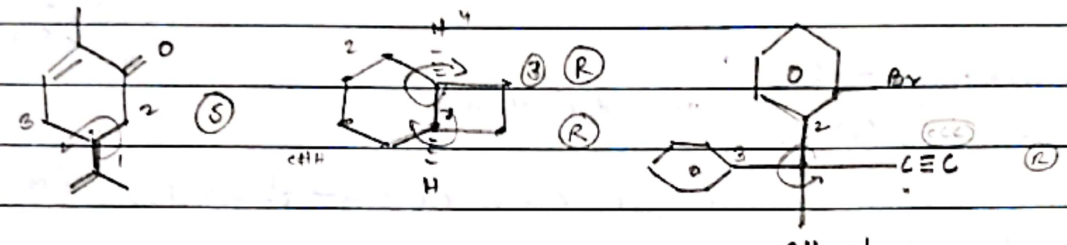
ACW (horizont) (R)

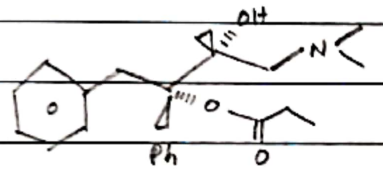
Q. 

} no chiral centre.

Q 

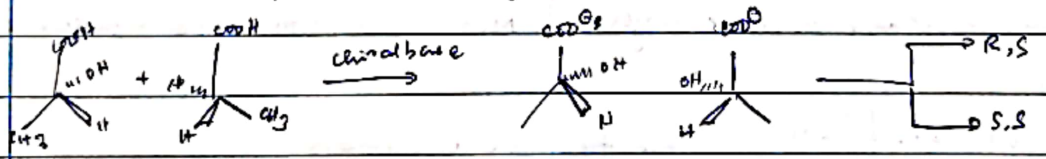
Q 

Q 

Q 

Racemic Mixture and Racemisation :

↳ 1:1 equimolar ratio of enantiomers → optically inactive.
they can be separated using simple diastereomerization process.



- Enantiomeric Excess : $ee\% = \dfrac{\text{specific rotation of sample}}{\text{specific rotation of pure}}$

Q. R[+] 2Bromo-butane $\alpha = 25°$ } $[\alpha]$ mix = 12.5. Find ee% and R%.
ee% = 50 and $x + (x+50) = 100 \Rightarrow x = 25\%$ } Hence R% = 75%.

Q. $[\alpha]$ of (+)X isomer is 35° and observed $[\alpha]$ sample −13.5°.
ee = 38.6% } 69.3 (−) isomer and 30.7 (+) isomer.

Q. [α] S(−) glyceraldehyde −8.7°. Observed rotation = −1.4 } %

ee → 16% ⇒ 58 and 42% → S and R

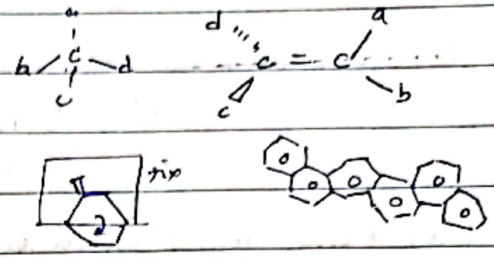
- Special Causes of Chirality

→ point chirality : chiral centers.

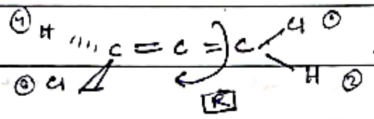
→ chiral axis : cis/trans.

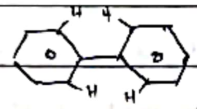
→ Chiral plane : keep one const, rotate

→ helical chirality : helicenes.



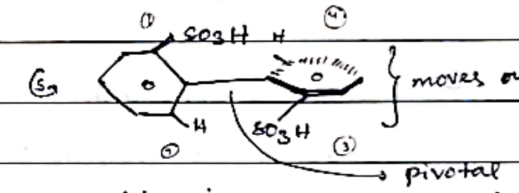
conjugated isolated c=c=c=c cumulative.

↳ if we have  ← Away and clockwise.

↳  extended conjugated structure.

↳ They keep on rotating, hence we cannot separate.

when we substitute the H with a bulky group, one of them moves away → being O.A.



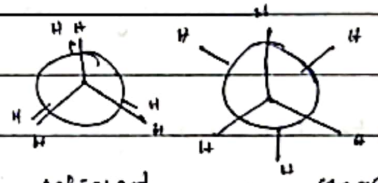
Atropisomers.

- Conformational Analysis:

stablest configuration = conformer.

– steric repulsion

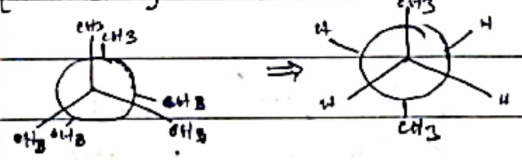
– bond opposition repulsion.



eclipsed staggered.

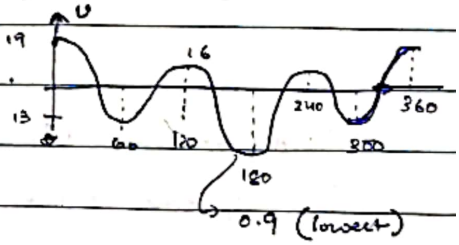
[Ethane].

[n-butane]



(most stable) → anti config prevents any steric.

↳ Draw configurations by rotating 60° multiple times.



Conformational Analysis of Cyclic Systems:

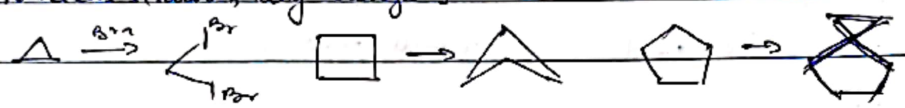
Angle Strain

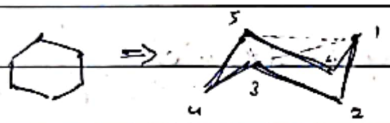
Now we know all the atoms here are sp^3 hybridized. So the deviation from act.

$= \frac{|\text{Actual} - \text{Observed}|}{2}$ } From here we got stability order: c. pentane > c. hexane > c. butane > c. propane.

↳ Later it was found cyclohexane is more available.

To release strain, they change form:

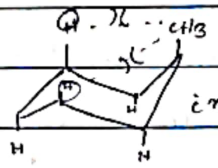




(chair form is usually the most stable).

- See that 1-3-5 are in one plane and 2,4,6 are in another plane.
- axials (a) → vertical || equatorial (e) → should be parallel to next c-c bond.

↳ Methyl Cyclohexane:



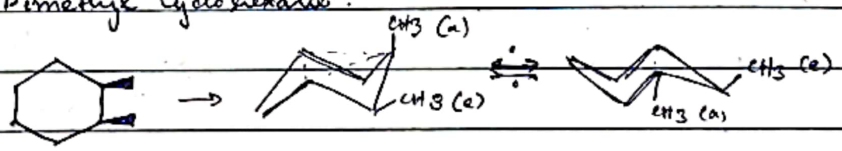
1-3 diaxial interaction ⇒ 0.9 kJ/mol ↑ interaction.

so, to decrease the stability, ring flip occurs.

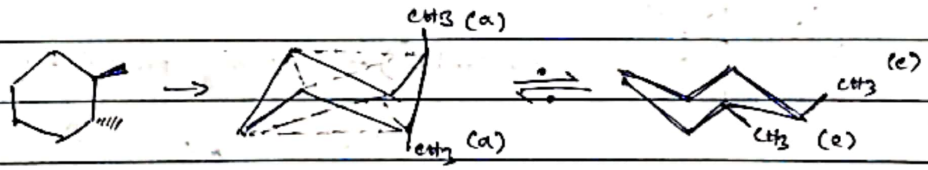
: basically axial ⟷ equatorial interchange.

CH3 (e).

↳ Dimethyl Cyclohexane:



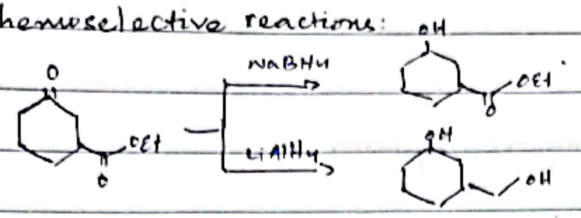
} The diaxial interaction still exists.



} much more stable.

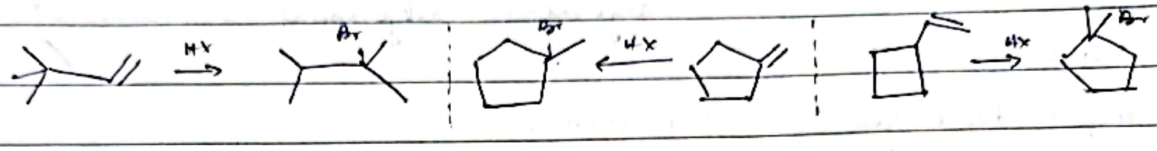
- Aspartame: (S,S) configuration → sweet whereas (R,S) is detected by bitter sensor instead.
- Cisplatin → anticarcinogen whereas transplatin is very toxic.

Chemoselective reactions:

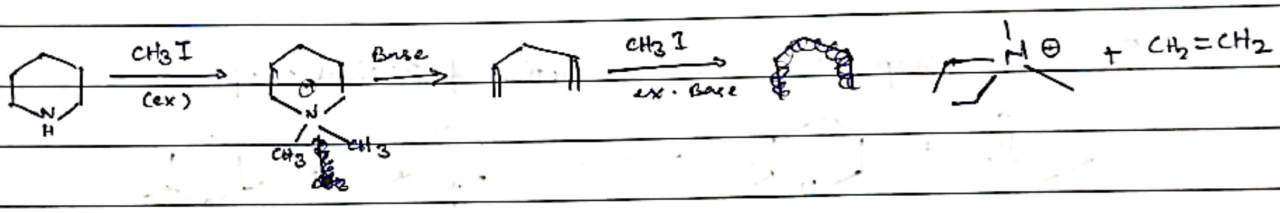


which of the reaction site react without reagent?

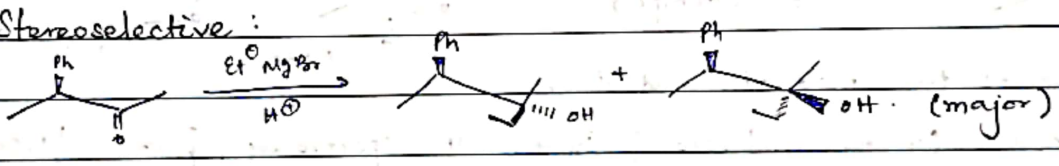
regioselective → decides where? Stereoselective → decides how?

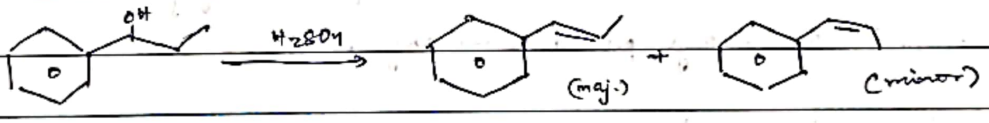


↳ when we have a bad (steric) leaving group, then less subs. product is formed



Stereoselective:





Photochemistry of Organic Molecules:

$A \xrightarrow{h\nu} A^* \longrightarrow$ photochemical } or photophysical.

$\quad\quad\quad\quad \xrightarrow{\Theta} A + B^*$ } energy transfer.

Relaxation by release of heat / light : photophysical change.

$S_0 \longrightarrow S_1$ = excitation.

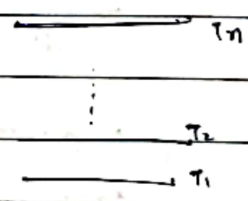
$S_n \longrightarrow S_1$ } vibrational relaxation.

$S_1 \longrightarrow S_0$ + light } Fluorescence

$S_1 \longrightarrow S_0$ + heat } Internal conversion

$S_1 \longrightarrow T_n$ ⇒ intersystem crossing

$T_n \longrightarrow S_0$ + light ⇒ phosphorence.

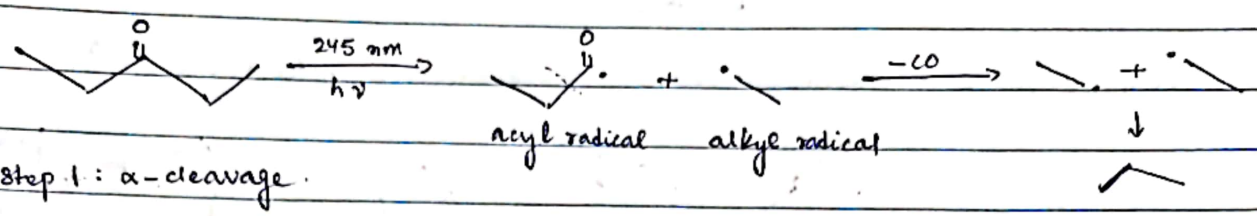


Date: __.__.__

Norrish experiment:

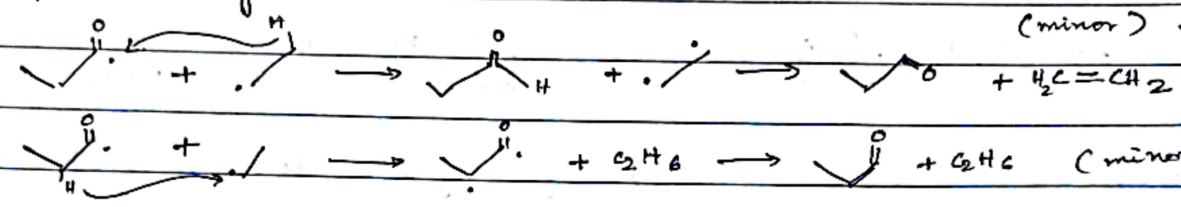
Type-I (α-cleavage reactions)

- can be observed in cyclic and acyclic systems.

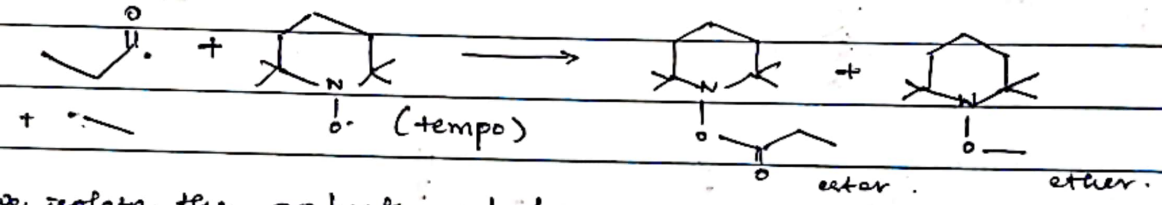


Step 1 : α-cleavage.

Step 2 : decarboxylation + recombination.



To confirm step 1 : we used the method of radical trap.

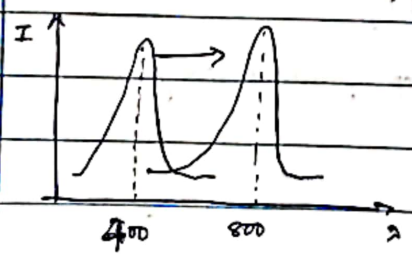


↳ we isolate the products and observe two peaks (of ether and ester) in the IR spectroscopy } 1730 | 1650 nm.

tempo = 2,2,6,6 - tetramethyl piperidine-1-oxyl.

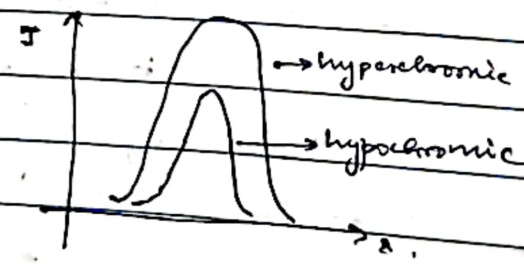
Q.

Electron rich : ε ↓ λ ↑ || less electron rich ε ↑ λ ↓



bathochromic shift : λ ↑ (moves towards higher)

hypsochromic shift : λ ↓ (moves towards lower)



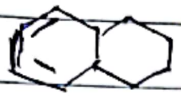
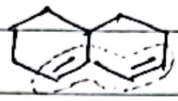
Hyperchromic → higher intensity.

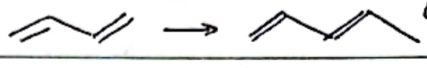
hypochromic → lower intensity.

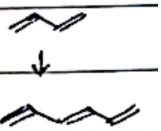
↳ we take a reference with solvent, while doing spectroscopy.

↳ saturate molecules don't absorb UV light.

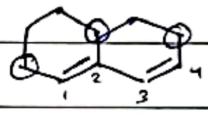
→ 217 nm. || Woodward – Fischer

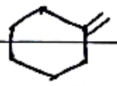
 Homoanular → 213 nm.  Heteroanular → 214 nm.

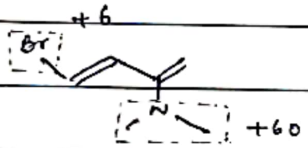
- Alkyl groups generally shift by +5 nm. }  (222 nm).

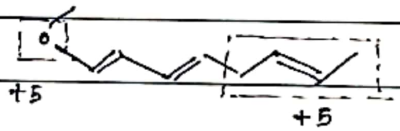
- N< : +60 nm. S< : +30 nm. } Halide : +6 nm. double bond : +30 nm. 

} : 0 nm (esters) for ethers : + 5 nm.

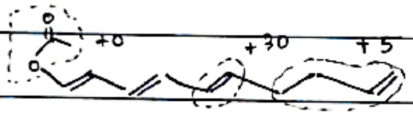
- Ring Residues : $sp^3 \rightarrow sp^2 \rightarrow$ part ring. RR : +5 nm.  3 RRs.
at each sp^2 check if its connected to a sp^3.

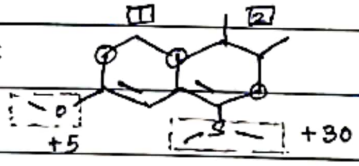
- Exo and Endo double bonds and 
↳ +5 nm ↳ (not mentioned) → 0. endo exo

Q: 

Q: 

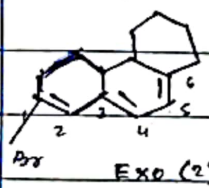
alkyl only since not in conjugation.

Q 

Q: 

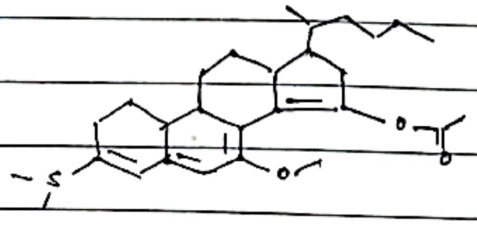
3 RRs || +15
For ring 1, dB is outside so exo double bonds +5nm

We only consider substituents that are present on sp^2 carbons.

Q. 

Exo (2)

Homo → 253
Br (+6).
RR (4×5 = 20) ⇝ conjugated (+30)

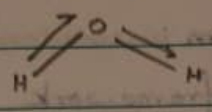
Q. 

Woodward Rules :
Applicable for $n<4$ double bonds.

Date: __/__/__

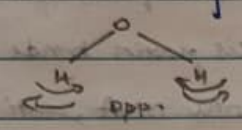
IR Spectroscopy (Functional Groups)

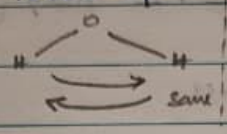
↳ Stretching : uniform | non-uniform

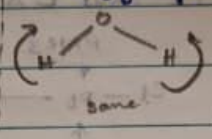
(symmetric)  (asymmetric)

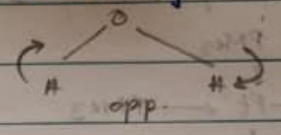
↳ bending vibration :

scissoring | rocking | wagging | twisting









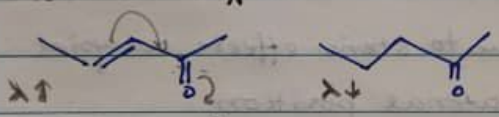
Stretching vibration > bending vibration energy ; IR Freq ↑
(3500 cm^{-1} – 1600 cm^{-1}) (1600 cm^{-1} – 600 cm^{-1})

$$\nu = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}$$

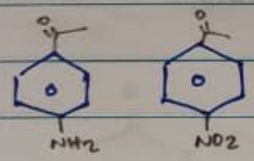
k → Force const (bond strength)
μ → reduced mass

} Hooke's Law

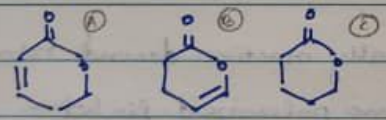
① Electronic Effect :

Q. 

since conjugation increases stability
↑stability ⇒ less energy.

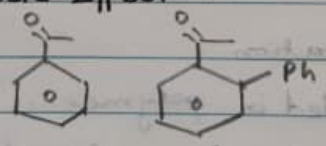
Q. 

} NH_2 will participate in resonance
DB gets weaker.

Q. 

Stretching Freq: B > C > A

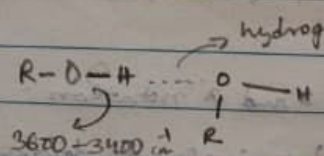
② Field Effect :



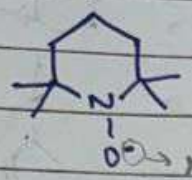
Ph will cause steric crowding for which it'll go to different plane and hence db ↑ν ↑

③ Hydrogen Bonding :

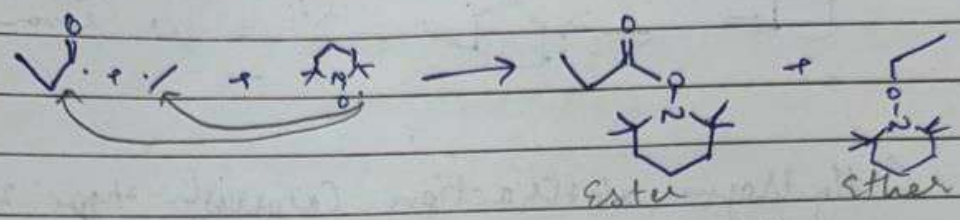
when there's hydrogen bonding the stretching freq ↓ hence non-polar mein higher.



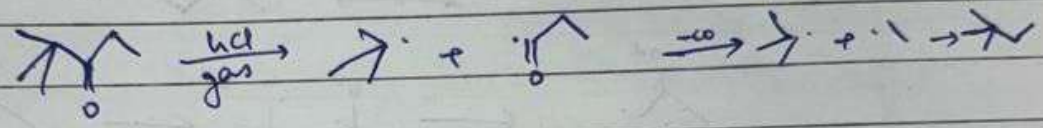
Radical trap :



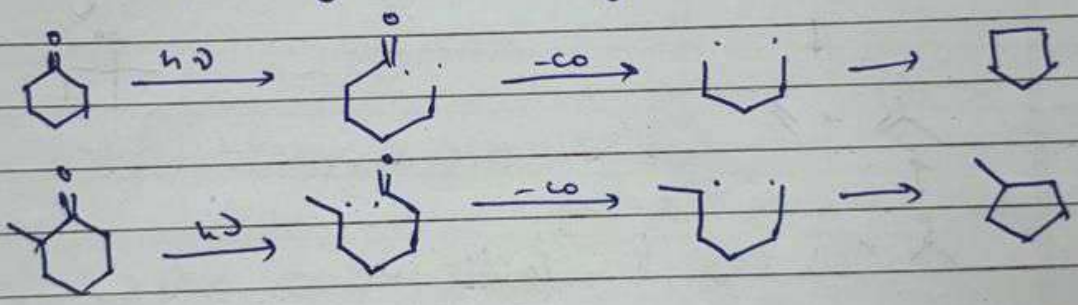
TEMPO : 2,2,6,6 - Tetramethyl pipiridine oxyl radical



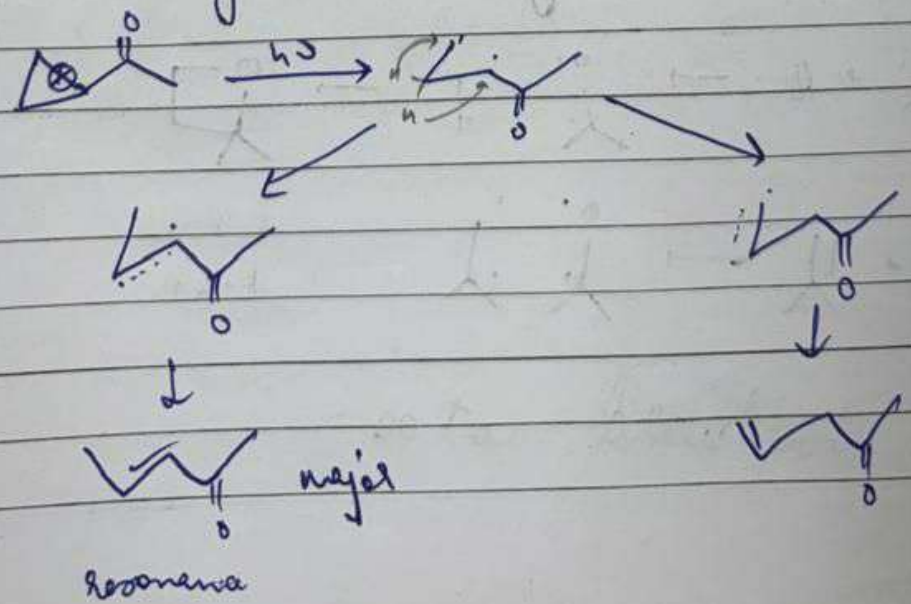
Can observe diff prdt by spectroscopy (diff wavelength for ester & ether)

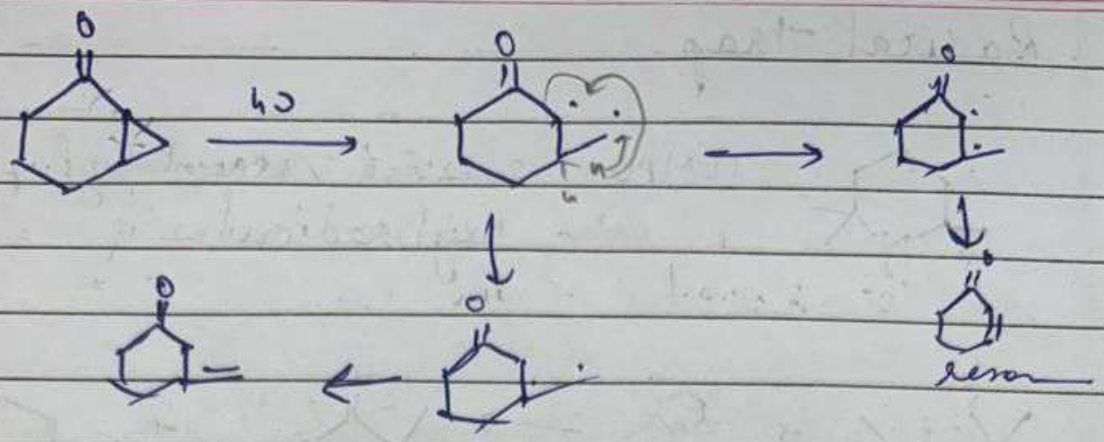
Eg 

* Norrish Type - 1 (Acyclic)

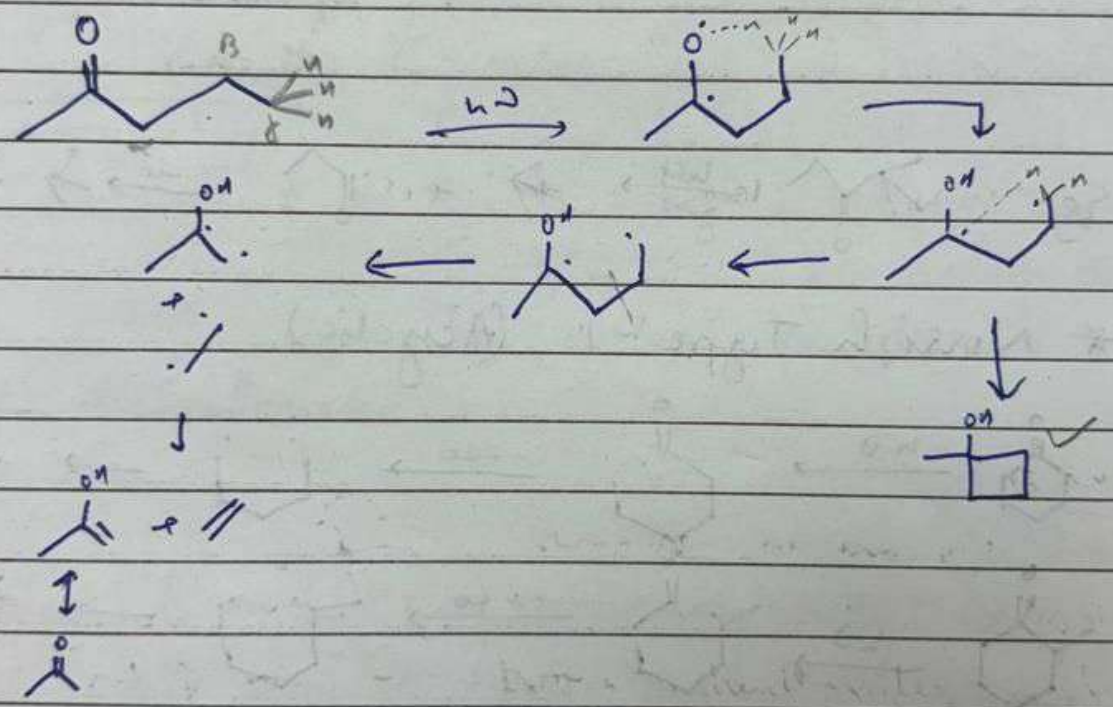
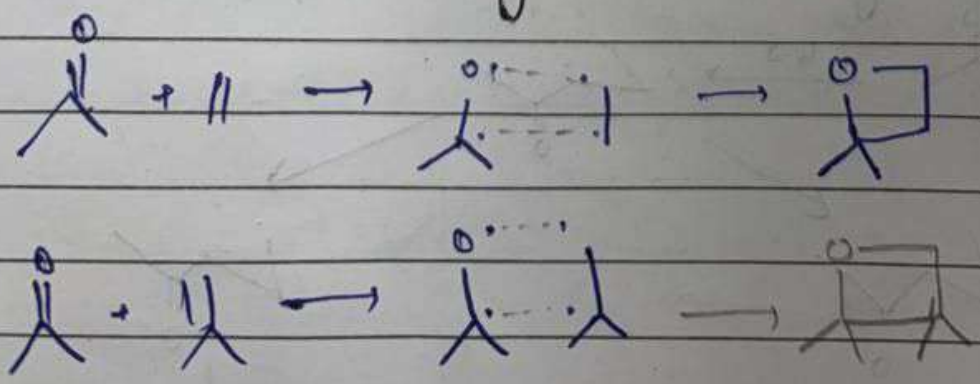


β cleavage :





* γ-Hydrogen abstraction (Norrish type 2)



Photochemical cyclo-addition rxn'